

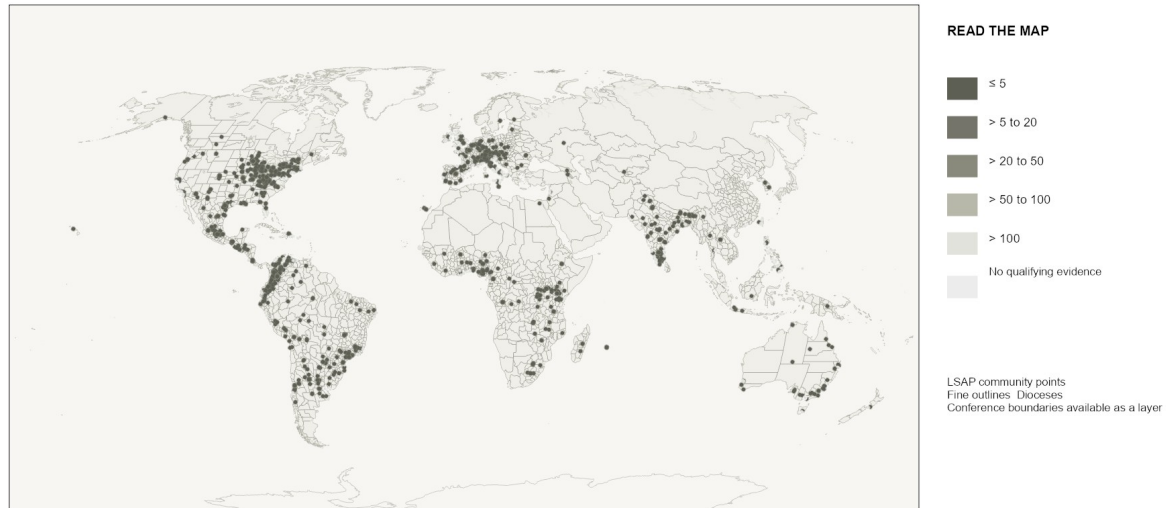
LSAP communities near river plastic outfalls

This report explains the source data, executed workflow and interpretation of lsap communities near river plastic outfalls. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Meijer et al., 2021, data set and article).

Map P07 | Documentation date 6 October 2026 | Units Kilometres | Model published 2021

LSAP communities near river plastic outfalls

Nearest river outfall · geodesic distance in km



Meijer et al. 2021 · copied LSAP member foundations

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P07

Figure P07 World overview exported from the completed ArcGIS Pro layout.

Proximity is straight-line geographic distance. A nearby outfall may belong to another watershed.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Meijer et al., 2021, data set and article).

1. Copy LSAP points into Member_Plastic_Proximity; preserve foundational member fields.
2. Generate geodesic nearest-outfall distances; join outfall identifier and modelled annual emission.
3. Summarize outfalls within 20 km and 100 km for every member.
4. Symbolize nearest distance at 5, 20, 50 and 100 km; nearby members receive darker pollution colors. Retain all dioceses and optional conference boundaries.
5. Create member popups with nearest distance, emission and counts.

Output fields and files

Field or band	Meaning
NearOut_ID and Near_km	Nearest outfall identifier and geodesic distance
Near_t_yr	Modelled emission of nearest outfall
Outfalls20 and Outfalls100	Outfall counts within 20 and 100 km
DistanceNote	Interpretation and positional precision note

Analysis dataset: PollutionAnalysis.gdb/Member_Plastic_Proximity. Styles/P07.lyrx and Styles/P07.mapx preserve layer and map definitions. Map_Exports/P07_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Distances use supplied member coordinates; positional precision has not been independently verified. Nearby outfalls may drain another watershed.

The Meijer midpoint GIS data are CC BY 4.0. Retain the authors, data DOI, scenario and units with any reused source or derived map. The original shapefile and copied outfall layer remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Meijer, L., van Emmerik, T., van der Ent, R., Schmidt, C., & Lebreton, L. (2021). Supplementary data for More than 1000 rivers account for 80 percent of global riverine plastic emissions into the ocean (Version 1) [Data set].

Figshare. Midpoint emissions shapefile. [Source link](#)

Meijer, L. J. J., van Emmerik, T., van der Ent, R., Schmidt, C., & Lebreton, L. (2021). More than 1000 rivers account for 80 percent of global riverine plastic emissions into the ocean. *Science Advances*, 7(18), eaaz5803. [Source link](#)

Esri. (n.d.). Generate Near Table. ArcGIS Pro documentation. Reference for the geodesic proximity tools used in this project. Accessed 6 October 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.